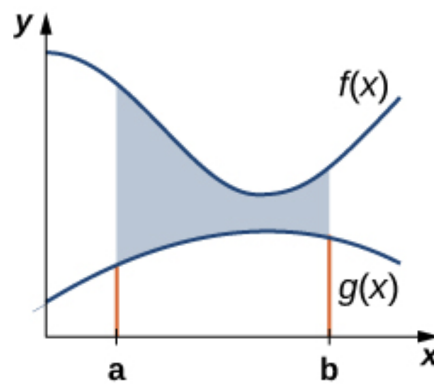
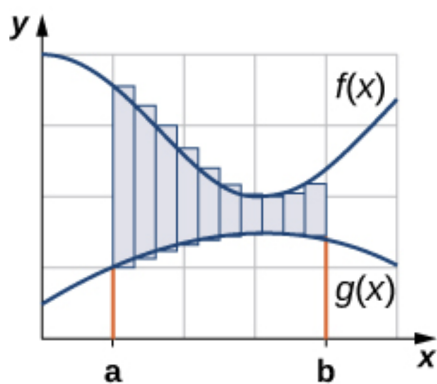


Section 2.1: Areas Between Curves

Definition 8. The area of the region between curves f and g with $f(x) \geq g(x)$ on $[a, b]$ is

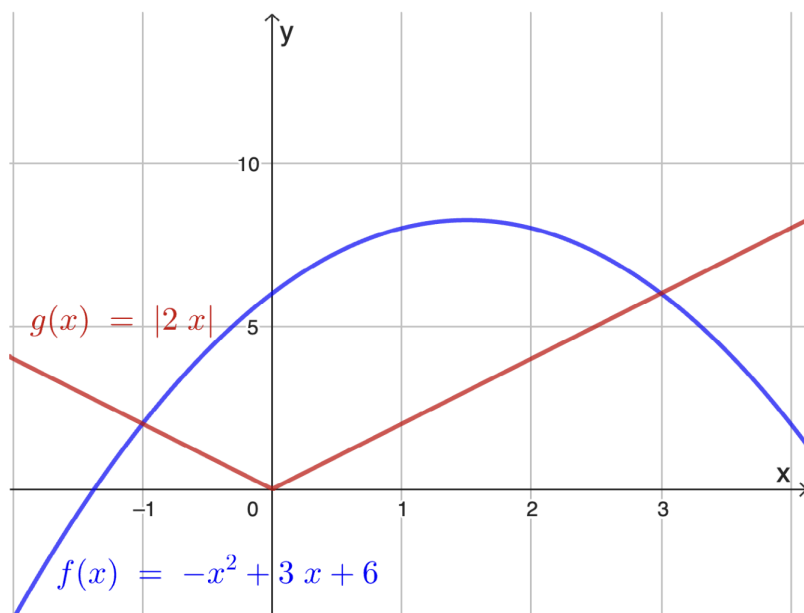
$$A = \int_a^b f(x) - g(x) dx$$

Visually. Discrete to continuous:

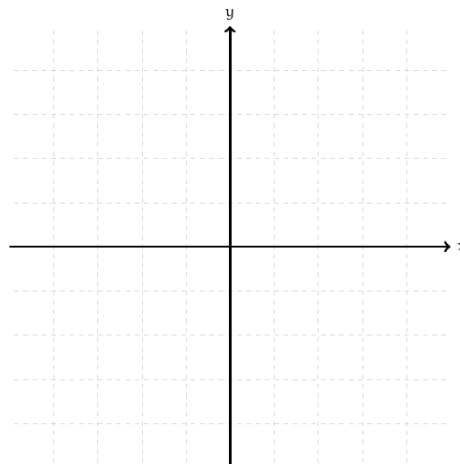


Link to interactive: <https://www.geogebra.org/m/ya2X7MAJ>

Example 74. Set up (but do NOT calculate) the integral needed to determine the area enclosed by the two graphs. Remove all absolute values.

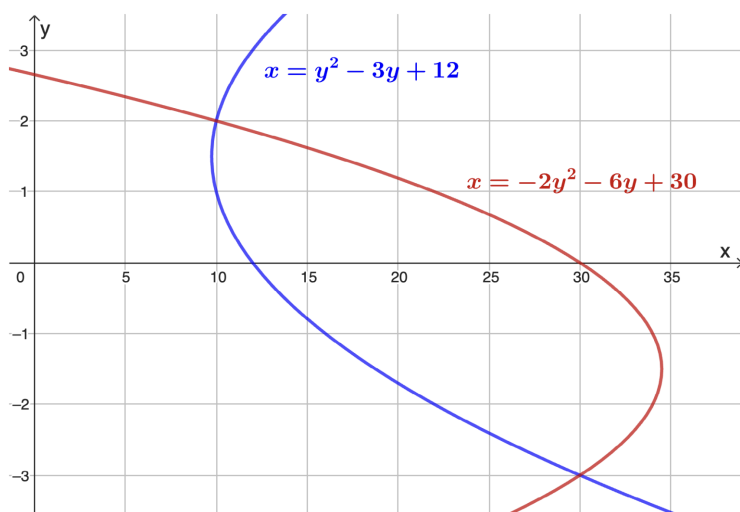


Example 75. Set up the integral(s) needed to find the area of the region bounded by $f(x) = 3x - x^2$ and $g(x) = -x$.



Example 76. Set up the integral(s) needed to find the area of the enclosed region between the curves defined by

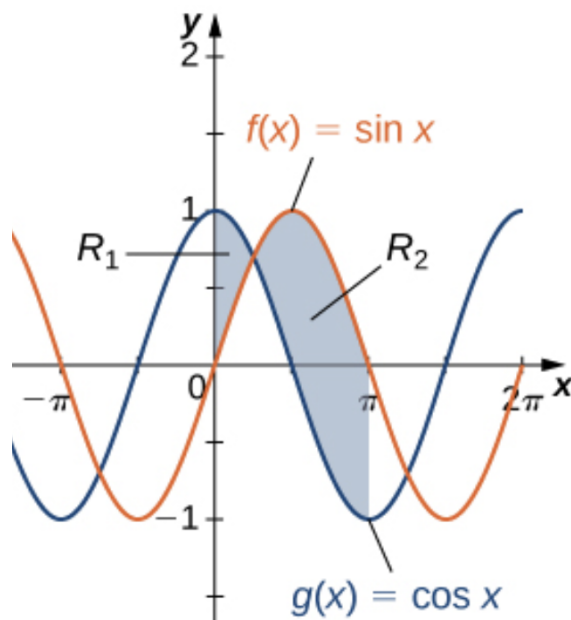
$$3y + x - 12 = y^2 \quad \text{and} \quad x + 2y^2 = 30 - 6y.$$



Theorem 16. For integrable functions f , g , the area of the region between the two curves on $[a, b]$ is the area.

$$A = \int_a^b |f(x) - g(x)| dx$$

Example 77. Set up the integral(s) needed to determine the area of the region between the graphs of $f(x) = \sin(x)$ and $g(x) = \cos(x)$ on $[0, \pi]$.



Example 78. Reflection. Discuss pros and cons of attempting the previous example by integrating with respect to y .

Example 79. Optional problem.

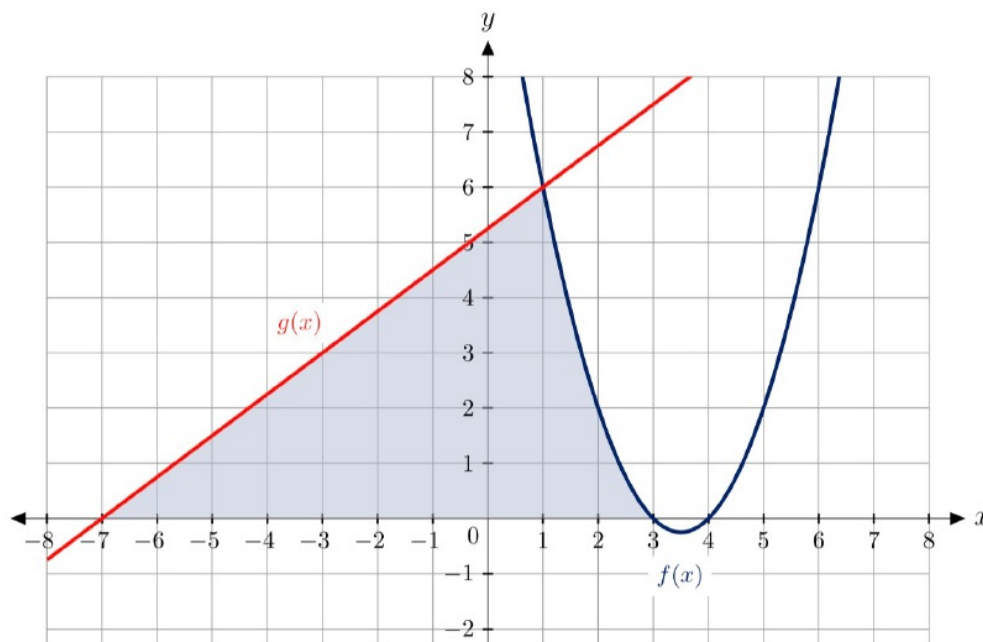
The tortoise versus the hare: The speed of the hare is given by the function $H(t) = -7t^2 + 38t$ whereas the speed of the tortoise is $T(t) = 3t$, where t is time measured in hours and the speed is measured in miles per hour. Find the distance between the tortoise and the hare, or the area between the curves, from the first time they are traveling at the same speed until one hour after. Enter answer using an exact value.

miles

Example 80. Optional problem. Set up ONLY.

Given the functions $f(x) = x^2 - 7x + 12$ and $g(x) = \frac{3x}{4} + \frac{21}{4}$, write the integral to find the area of the region pictured below that is bounded above by $f(x)$ and $g(x)$ and bounded below by the x -axis when integrating with respect to y .

Do not evaluate the integral.



Hint for above: Use the **quadratic formula** to solve for $x = \dots$

$$y = x^2 + bx + c \implies 0 = x^2 + bx + (c - y) \implies x = \frac{-b \pm \sqrt{b^2 - 4(1)(c - y)}}{2}$$

1. Left leg of the parabola: $x = \frac{-b - \sqrt{b^2 - 4(c - y)}}{2}$

2. Right leg of the parabola: $x = \frac{-b + \sqrt{b^2 - 4(c - y)}}{2}$

Section 2.2a: Determining Volumes by Cross-Sectional Slicing

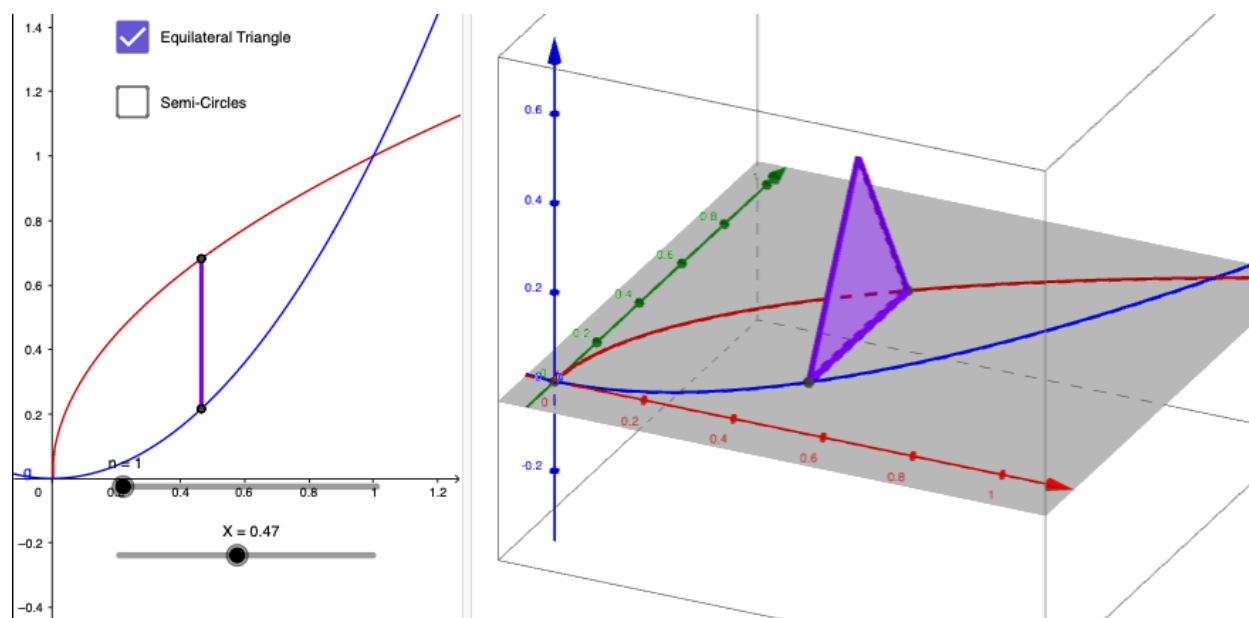
Definition 9. If S is a solid with cross sections between $x = a$ and $x = b$ with area function $A(x)$, which is continuous, then

$$\text{Volume of } S = \int_a^b A(x) dx.$$

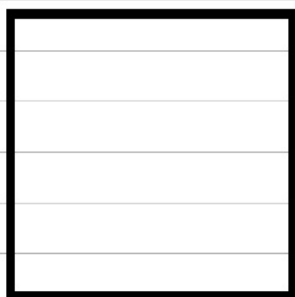
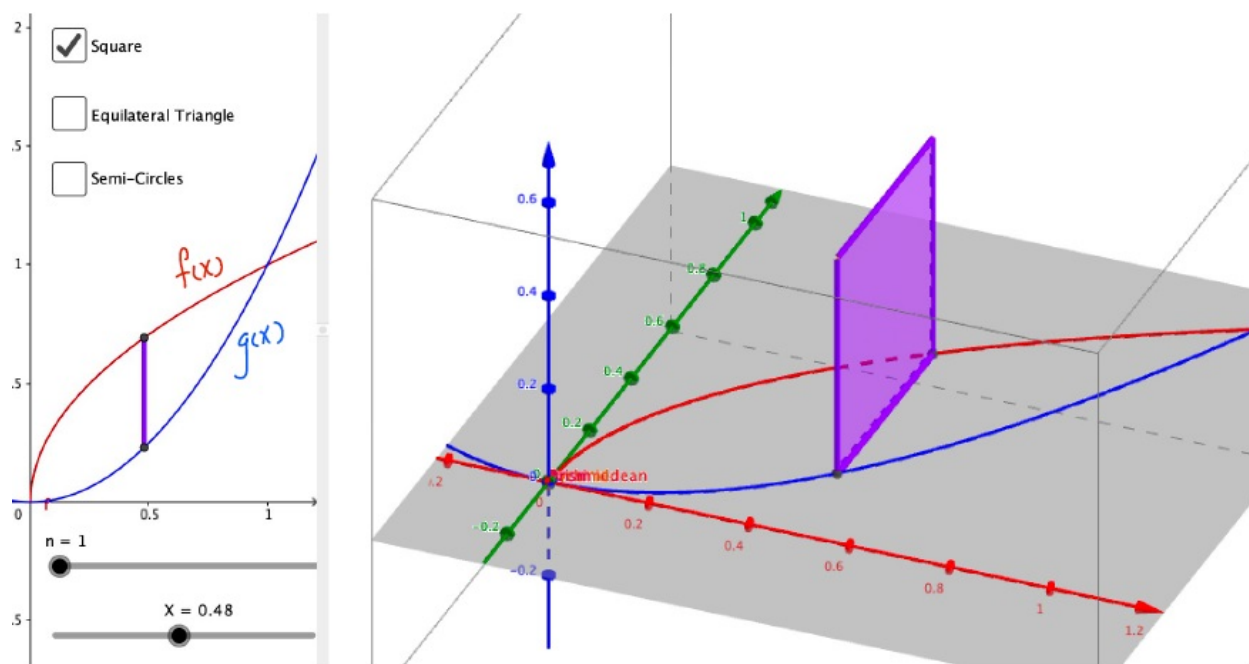
Vocabulary: Method of **cross-sectional slicing**.

Link to interactive: <https://www.geogebra.org/m/XFgMaKTy>

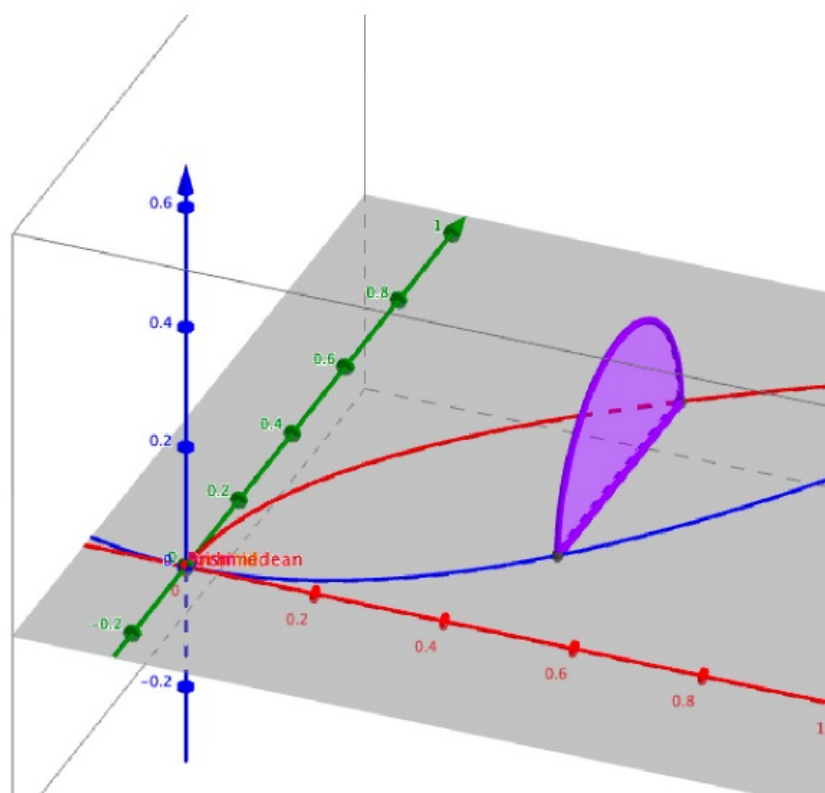
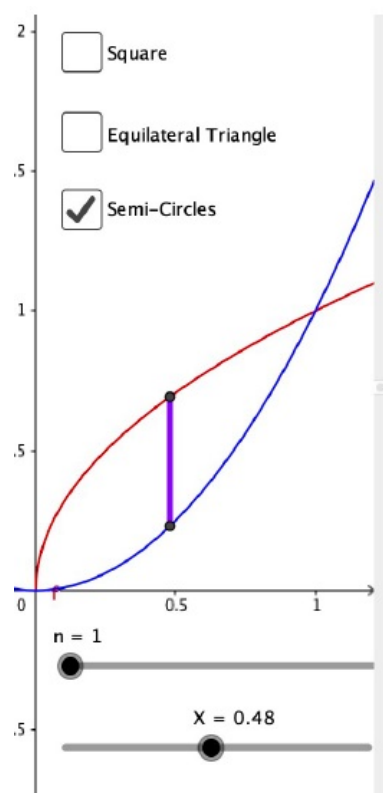
Example 81. Consider the three-dimensional cross-section. Set up the integral for cross-sectional area. Use the area formula for an equilateral triangle: $A(b) = \frac{\sqrt{3}}{4} \cdot b^2$



Example 82. Consider the three-dimensional cross-section. Set up the integral for cross-sectional area.



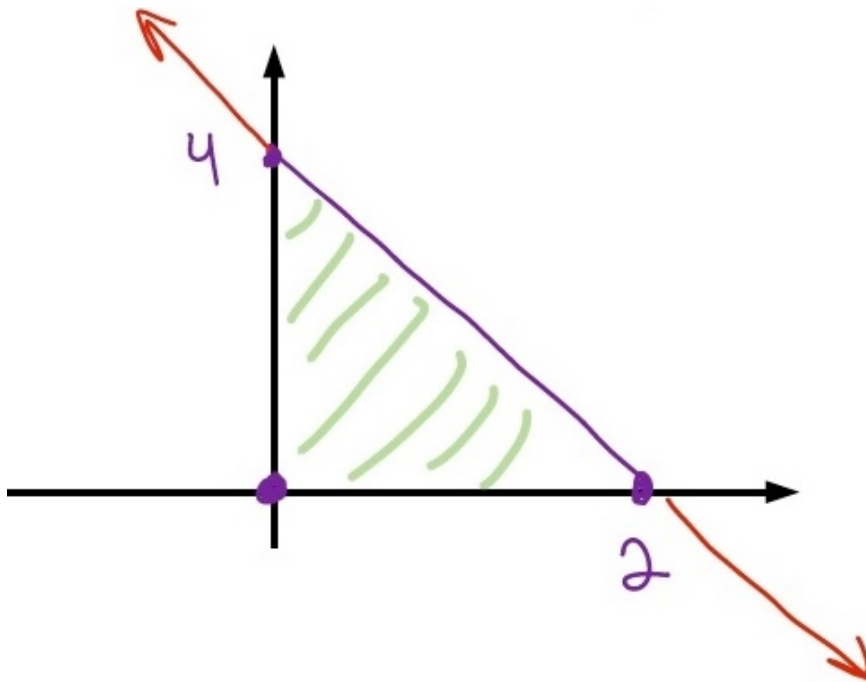
Example 83. Consider the three-dimensional cross-section. Set up the integral for cross-sectional area.



Example 84. Set up the integral using **cross-sectional slicing** to find the volume of the solid whose base is in the first quadrant bounded by $y = \sqrt{10 - 2x}$, the x -axis, and $x = 0$ using equilateral triangle cross-sections perpendicular to the x -axis.

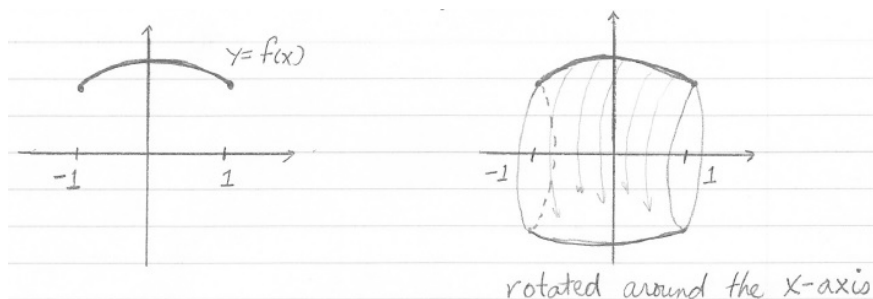
Hint. Use the area formula: $A(b) = \frac{\sqrt{3}}{4} \cdot b^2$

Example 85. Set up the integral using **cross-sectional slicing** to find the volume of the solid whose base is a triangle with vertices $(0,0)$, $(0,4)$, $(2,0)$. Slices perpendicular to the x -axis are semicircles.

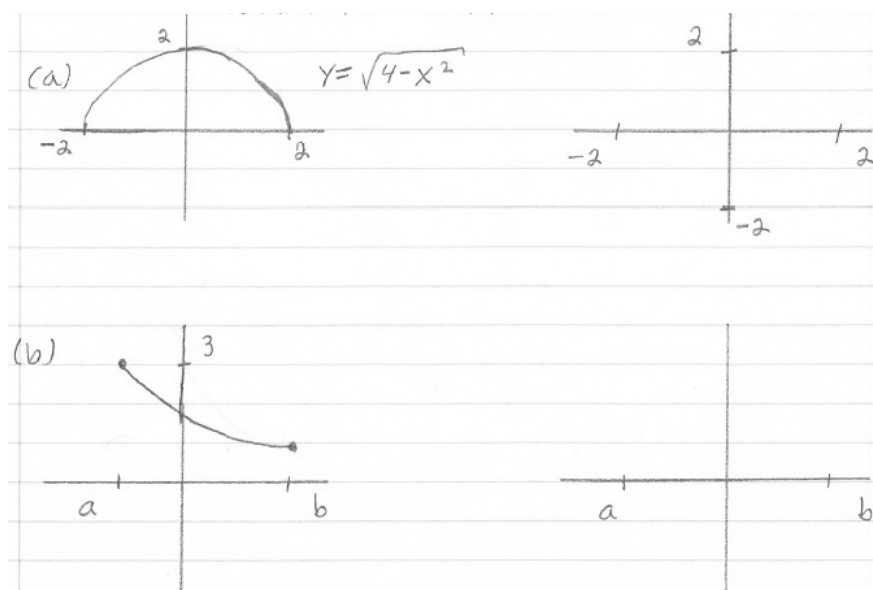


Section 2.2b: Disk/Washer Method

Consider the region bounded by $f(x) = \sqrt{4 - x^2}$ and the x -axis on $[-1, 1]$.



Example 86. Sketch the final figure after rotating about the x -axis.



Definition 10. A solid of revolution is a 3-dimensional object obtained by rotating a planar, 2-dimensional object around an axis.

Definition 11. Set S be a solid of revolution on $[a, b]$ around the x -axis with radius $r(x)$, then

$$\text{Volume of } S = \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi r(x_i^*)^2 \Delta x = \pi \int_a^b r(x)^2 dx$$

Sketch Idea:

Link to interactive: [Disk Method Link](#)

Definition 12. Volume of washers: (Outer revolution) – (inner revolution)

Volume of washer $S = \pi \int_a^b R(x)^2 - r(x)^2 dx$

Sketch Idea:



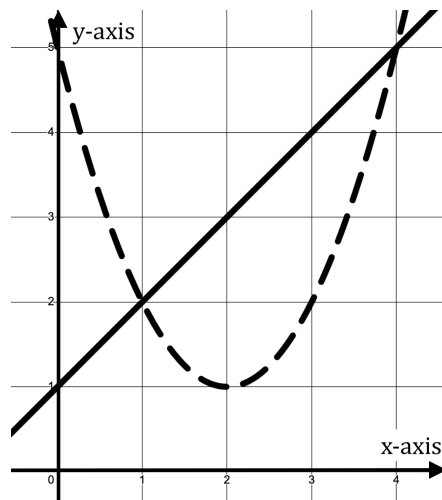
Link to interactive: <https://www.geogebra.org/m/fy3c4pyf>

Example 87. (Disk Method) Using the Disk/Washer Method, set up the integral(s) needed to compute the volume of the solid formed when rotating around the x -axis the region bounded between the two graphs of: $r(x) = \sqrt{4 - x^2}$ and $y = 0$ (or the x -axis).

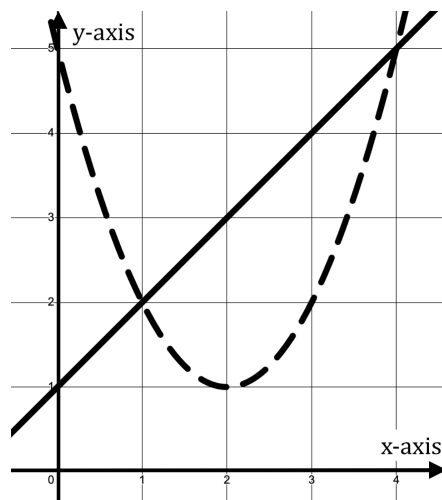
Example 88. (Washer Method) Using the Disk/Washer Method, set up the integral(s) needed to compute the volume of the solid formed when rotating around the x -axis the region bounded between the two graphs of: $r(x) = \sqrt{4 - x^2}$ and $y = \sqrt{3}$ on $[-1, 1]$.

Example 89. Using the Disk/Washer Method, set up the integral(s) needed to compute the volume of the solid formed when rotating around the x -axis the two regions bounded between the graphs of: $f(x) = x + 1$ and $g(x) = x^2 - 4x + 5$ and $x = 0$ on $[0, 4]$.

Verify the intersection points.



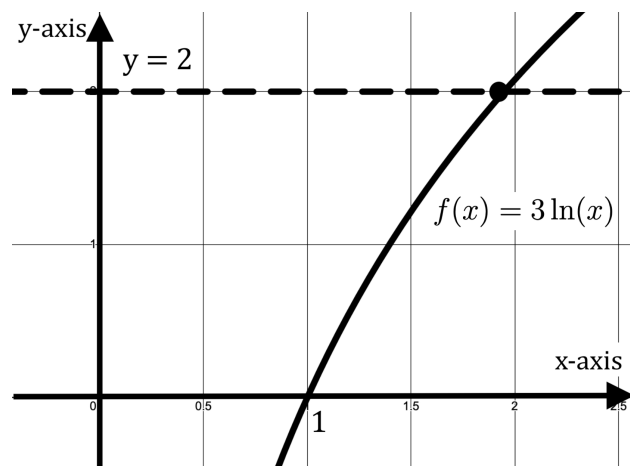
Example 90. Using Example 78 (above), set up (but to not evaluate) the integral to find the volume if the regions were revolved around the horizontal axis $y = -3$.



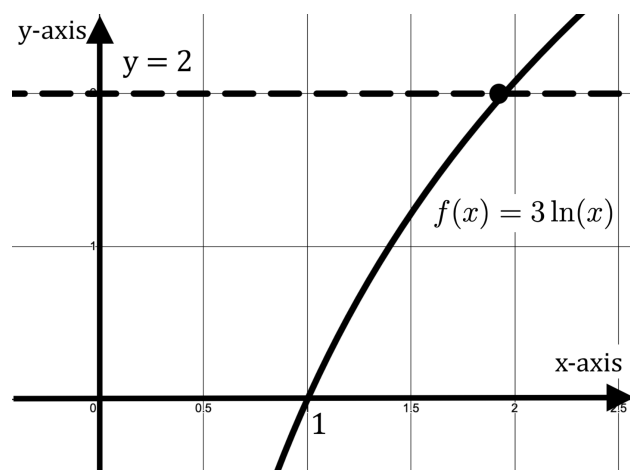
Example 91. Using the Disk/Washer Method, set up the integral(s) needed to compute the volume of the solid formed when rotating the region bounded between the graphs of: $f(x) = 3 \ln(x)$, the line $y = 2$, and the x - and y - axes.

Verify the intersection points.

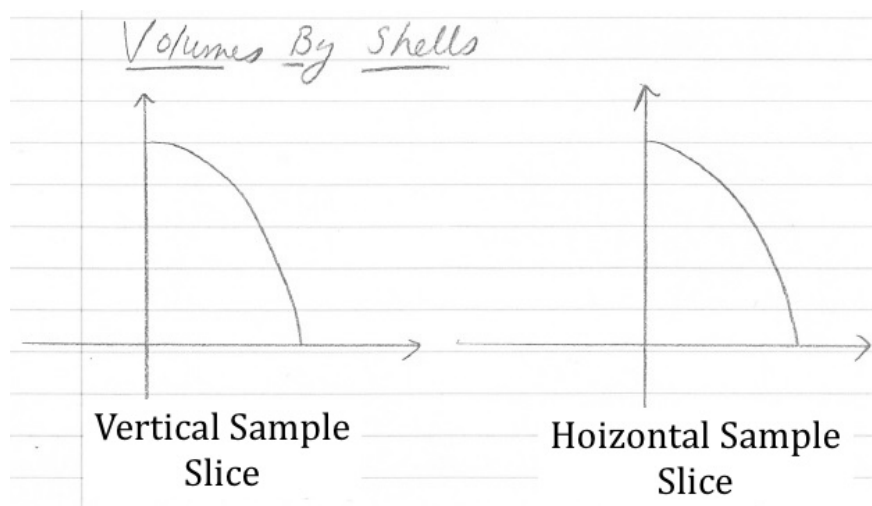
1. Rotating around the line $y = 3$.



2. Rotating around $x = 0$ (the y -axis).



Section 2.3: Volumes by Shells



Theorem 17. A shell with inner radius r_1 , outer radius r_2 , and height h is said to have average radius $r = \frac{r_1+r_2}{2}$ and thickness $\Delta r = r_2 - r_1$. The volume is $V = 2\pi r h \Delta x$.

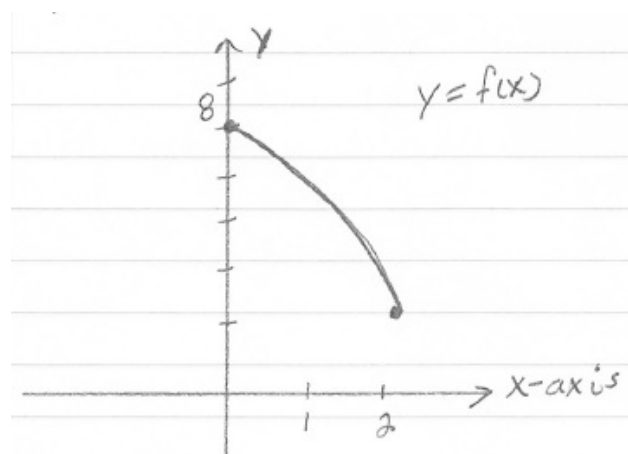
Definition 13. A solid of revolution obtained by rotating a region on $[a, b]$ around the y -axis has

$$\text{Volume of solid with nested shells} = \lim_{n \rightarrow \infty} \sum_{k=1}^n 2\pi r_k h_k^* \Delta x = 2\pi \int_a^b r(x) h(x) dx$$

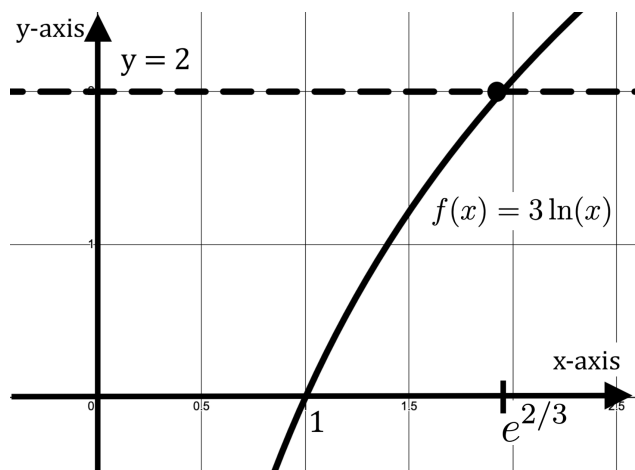
with $r(x)$ the average radii, and $h(x)$ the height.

Link to interactive: <https://www.geogebra.org/m/v7ymgg9x>

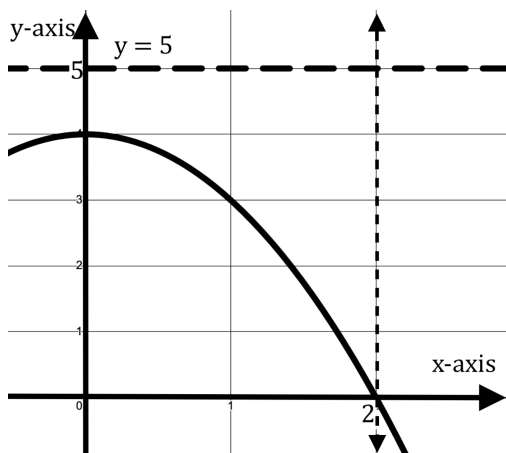
Example 92. (Shell Method) Find the volume of the solid obtained by rotating the region between the graph of $f(x) = -x^2 - x + 8$ and the x -axis on $[0, 2]$ around the y -axis.



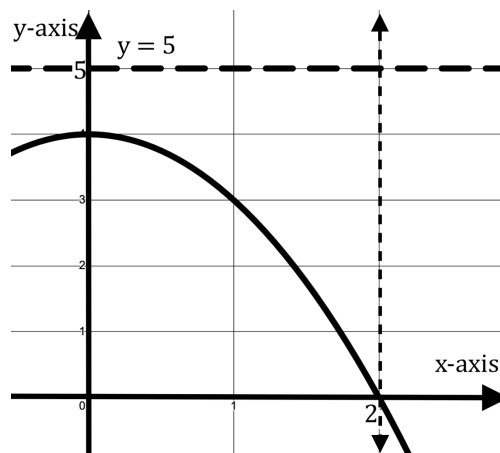
Example 93. Set up the integral to find the volume of the region bounded by the graph $y = 3\ln(x)$, the line $y = 2$, and the x - and y - axes, rotated around the x -axis.



Example 94. Using the Shell Method, set up the integral(s) needed to compute the volume of the solid formed when rotating the region bounded between the graphs of: $f(x) = 4 - x^2$ and $y = 5$ on $[0, 2]$ use the axis of rotation stated below.



(1) Around the y -axis.



(2) Around $x = 3$

Example 95. Define R as the region bounded by the graphs of $f(x) = 4\sqrt{x}$ and $g(x) = x^2$ on the interval $[0, 2]$. Which of the following represents the volume of the solid of revolution formed by rotating R about the line $x = 3$?

(A) $\int_0^2 2\pi(3-x)(x^2 - 4\sqrt{x}) dx$

(B) $\int_0^2 2\pi(3+x)(4\sqrt{x} - x^2) dx$

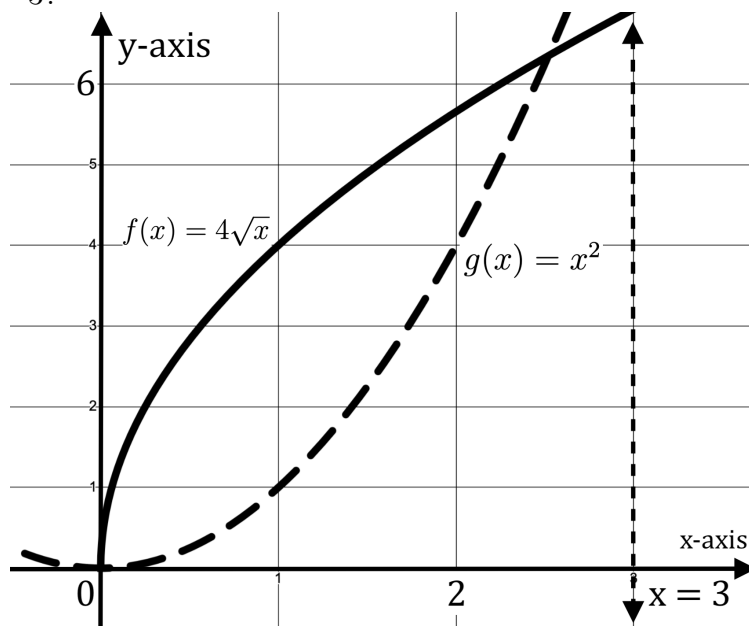
(C) $\int_0^6 2\pi(3-x)(4\sqrt{x} - x^2) dx$

(D) $\int_0^2 2\pi(x-3)(4\sqrt{x} - x^2) dx$

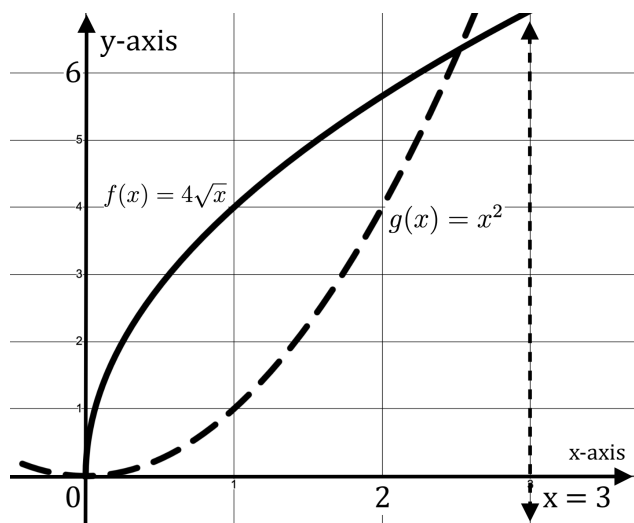
(E) $\int_0^2 2\pi(3-x)(4\sqrt{x} - x^2) dx$

(F) $\int_0^2 2\pi(3+x)(x^2 - 4\sqrt{x}) dx$

(G) $\int_0^3 2\pi(3-x)(4\sqrt{x} - x^2) dx$

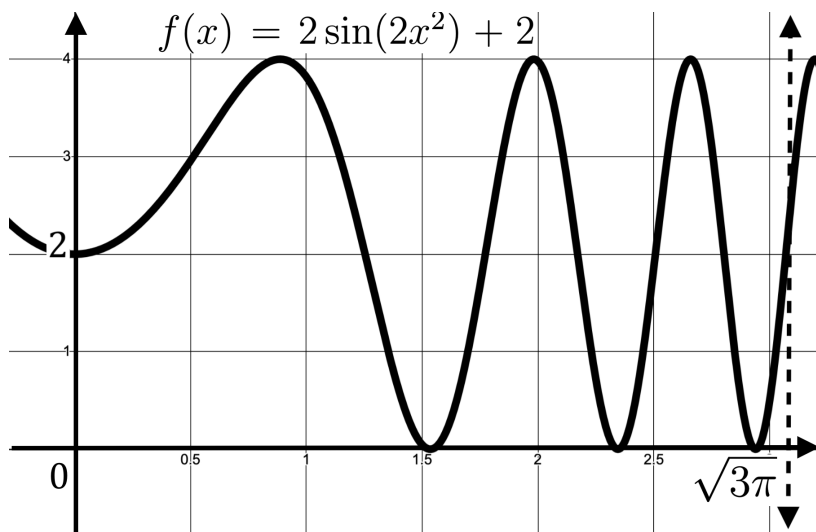


Optional Reflection: Discuss the pros/cons of using the Disk/Washer Method to compute the same volume as above.

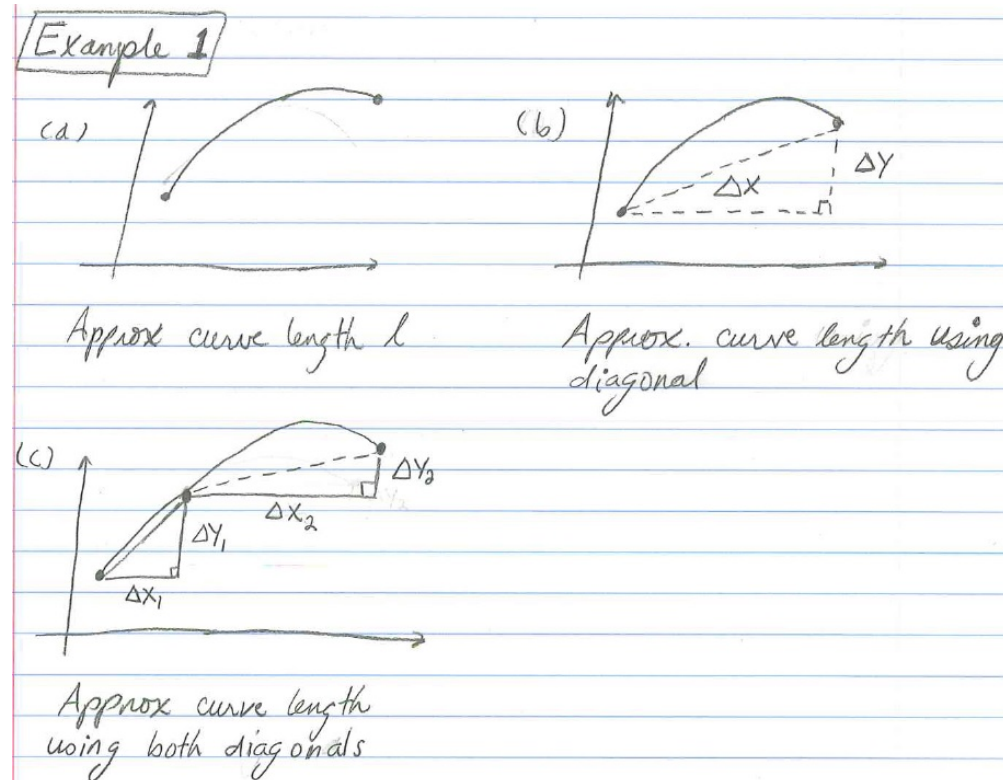


Example 96. (Optional) R is the region bounded above by the graph of $f(x) = 2 \sin(2x^2) + 2$ and below by the x -axis over the interval $[0, \sqrt{3\pi}]$. Find the volume of the solid of revolution formed by revolving R around the y -axis.

Submit an exact answer in terms of π .



Section 2.4: Arc Length and Surface Area



Total approximate arc length:

$$\begin{aligned}
 & \sum_{i=1}^n \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} \\
 &= \sum_{i=1}^n \sqrt{\left(\frac{\Delta x_i^2 + \Delta y_i^2}{\Delta x_i^2} \right) \cdot \Delta x_i^2} \quad (\text{Simplifying so } \Delta x_i = \Delta x \text{ are of uniform width}) \\
 &= \sum_{i=1}^n \left(\sqrt{\frac{\Delta x^2}{\Delta x^2} + \frac{\Delta y_i^2}{\Delta x^2}} \right) \cdot \sqrt{\Delta x^2} \\
 &= \sum_{i=1}^n \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x} \right)^2} \cdot \Delta x.
 \end{aligned}$$

Then taking the limit,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x} \right)^2} \cdot \Delta x =$$

Link to interactive: <https://www.geogebra.org/m/AQdFBK5N>

Example 97. Set up the integral needed to determine the arc length of the graph of $f(x) = 2\sin(2x^2) + 2$ over the interval $[2, \sqrt{3\pi}]$. Do Not Evaluate.

Example 98. Set up and evaluate the integral needed to determine the arc length of the graph of $g(y) = 4y - 2$ over the interval $[2, 7]$.

Example 99. Set up and evaluate the integral needed to determine the arc length of the graph of $h(x) = \frac{2}{3}(x + 1)^{3/2}$ over the interval $[0, 1]$.

Example 100. Let $g(x) = \ln(\cos(x))$. Calculate the arc length of the graph of $g(x)$ over the interval $[\pi/4, \pi/3]$.

Example 101. (Optional Challenge) Let $g(y) = \frac{y^3}{6} - 1 + \frac{1}{2y}$. Calculate the arc length of the graph of $g(y)$ over the interval $[1, 3]$.

Hint: Use algebra to create a perfect square.